A Project Report on

ResQFood – A Smart Platform to Reduce Food Waste and Support NGOs

submitted in partial fulfillment of the requirements for the award of the degree of

BACHELOR OF ENGINEERING

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CERTIFICATE

This is to certify that the project entitled "ResQFood – A Smart Platform to Reduce Food Waste and Support NGOs" is a bonafide work of **Vankeepuram Aarya Vardhan (24106007) , Waghe Sahil (24106013) , Yogi Ishwar (24106109) , Vyas Annanay (24106144)** submitted to the University of Mumbai in partial fulfilment of the requirement for the award of the degree of **Bachelor of Engineering** in **Computer Science & Engineering (Artificial Intelligence & Machine Learning**).

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Abstract

ResQFood is a technology-driven platform designed to reduce food wastage and address hunger by efficiently redistributing surplus food. In urban areas, a significant amount of edible food is wasted daily by restaurants, hotels, and event organizers, while many people continue to suffer from food insecurity. This project proposes a solution that connects food donors with NGOs and volunteers through a centralized system.

The platform allows donors to upload details of excess food, including quantity, type, and location. The system verifies the data and matches it with nearby NGOs based on proximity and requirement. Volunteers or delivery services are then coordinated to ensure timely pickup and distribution of food, supported by real-time tracking and notifications.

The proposed system aligns with Sustainable Development Goals (SDGs), particularly SDG 2: Zero Hunger, by improving access to food for the needy; SDG 12: Responsible Consumption and Production, by reducing food waste; and SDG 13: Climate Action, by minimizing environmental impact. It enhances coordination among stakeholders, promotes sustainability, and ensures surplus food reaches those in need efficiently.

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ABBREVATION

AI	*Artificial Intelligence*
ML	*Machine Learning*

Chapter 1

Introduction

1. Introduction

1.1 Overview

Food wastage is a significant global issue, especially in urban areas where large quantities of edible food are discarded daily by restaurants, hotels, and households. At the same time, many people suffer from hunger and lack proper nutrition, highlighting the need for an efficient redistribution system.

Existing food donation methods are mostly manual and lack proper coordination between donors and NGOs, leading to delays, mismanagement, and wastage of perishable food.

ResQFood is a technology-based platform designed to address this problem. It allows donors to upload surplus food details, matches them with nearby NGOs, and assigns volunteers for quick collection and delivery, with real-time updates to improve coordination.

1.2 Motivation

The motivation behind this project is to reduce food wastage and help people in need. A large amount of food is wasted daily due to lack of proper systems, while many individuals struggle to find basic meals.

ResqFood aims to use technology to solve this problem by creating a simple and efficient platform for food donation. It also helps improve coordination between donors, NGOs, and volunteers, making the process faster and more reliable.

Key motivations include:

- Reducing food wastage
- Supporting underprivileged communities
- Improving coordination among stakeholders
- Using technology for social impact

1.2.1 Scope

The scope of ResqFood includes managing the complete process of food donation and distribution through a centralized platform. Donors can upload food details, NGOs can accept donations, and volunteers handle delivery.

The system provides real-time notifications to ensure smooth communication and coordination. It is designed to be scalable and can be extended with additional features in the future.

The main scope includes:

- Food donation management
- NGO coordination
- Volunteer-based delivery system
- Real-time tracking and notifications

Chapter 2

Literature Survey/ Existing system

2. Literature Survey

Food waste management has gained significant attention in recent years due to its environmental, economic, and social impact. Various research studies and existing systems have attempted to address this issue using technology, logistics optimization, and community-driven approaches. This section reviews several existing solutions and research contributions related to food waste reduction and redistribution.

Many researchers have focused on developing digital platforms that connect food donors with recipients. These systems aim to reduce the gap between surplus food availability and demand by enabling real-time communication and coordination. However, challenges such as scalability, food safety, and efficient logistics still persist. Additionally, issues like lack of user awareness and inconsistent participation further limit the effectiveness of these solutions.

2.1 Review of Existing Systems and Research Papers

1. Food Rescue Applications (General Study)

Several food rescue applications have been developed to redistribute surplus food from restaurants and supermarkets to NGOs. These platforms primarily focus on listing available food and enabling volunteers to collect and distribute it. While they help reduce waste, many lack real-time tracking and efficient matching mechanisms.

2. "Food Waste Management System Using IoT"

This research proposes the use of IoT sensors to monitor food waste levels in bins and storage areas. The system helps in tracking waste generation and optimizing collection processes. However, it focuses more on waste monitoring rather than redistribution of edible food.

3. "Smart Food Redistribution System Using Web Technology"

This paper introduces a web-based platform that connects donors with receivers. It includes features such as user registration, food listing, and request handling. While effective, the system lacks advanced features like real-time notifications and location-based optimization.

4. "Reducing Food Waste Through Mobile Applications"

This study explores mobile-based solutions that allow users to donate surplus food. The application improves accessibility and convenience but faces challenges in ensuring timely pickup and maintaining food quality.

5. "Blockchain-Based Food Supply Chain Management"

This research focuses on using blockchain technology to ensure transparency and traceability in food distribution. While it enhances trust and accountability, it is complex to implement and may not be suitable for small-scale food donation systems.

6. "AI-Based Food Demand Prediction System"

This system uses machine learning algorithms to predict food demand and reduce overproduction. It helps minimize waste at the source but does not address the redistribution of already available surplus food.

7. "Online Food Donation System"

This project presents a basic platform where users can donate food and NGOs can request it. It improves communication but lacks automation, scalability, and real-time updates.

8. "Food Sharing Economy Platforms"

These platforms encourage individuals to share excess food within communities. While they promote sustainability, they often lack structured coordination and safety measures.

2.2 Comparative Analysis

From the above studies, it is evident that:

- Most systems focus either on waste monitoring or food redistribution, but rarely both
- Many platforms lack real-time communication and tracking
- Logistics and coordination remain major challenges
- Advanced technologies like AI and blockchain are used but are often complex and not widely accessible
- Existing solutions are often limited in scalability and usability

2.3 Summary of Literature Survey

The literature survey highlights that while several approaches have been proposed to tackle food waste, there is still a need for a comprehensive, user-friendly, and scalable solution. The existing systems either lack efficient coordination, real-time updates, or accessibility for common users.

The proposed system, ResQFood, aims to overcome these limitations by integrating modern web technologies, real-time communication, and location-based services to create a more effective and practical solution for food waste management.

Table 1:Analysis of Existing Systems

Platform	Core Model	Target Audience	Key Limitations
Too Good To Go	Market-based (Paid)	Restaurants to Consumers	Focuses on discounted sales rather than free donation; limited to commercial entities.
OLIO	P2P (Peer-to-Peer)	Individual Neighbours	Focuses on discounted sales rather than free donation; limited to commercial entities.
FoodCloud	B2B (Business-to-NGO)	Supermarkets to Charities	Requires high logistical coordination and large-scale warehouse storage.
ResQ Club	Location-based Sale	Commuters/Urbanites	Geographic limitation; primarily operates in European "Smart Cities."

Chapter 3

Limitation of Existing system

3. Limitation of Existing System

Despite the availability of various food waste management and redistribution systems, several limitations still exist that reduce their effectiveness. These systems often fail to provide a complete and scalable solution to food wastage and hunger.

One major limitation is the lack of real-time coordination between food donors and recipients. Many platforms do not support instant updates or notifications, leading to delays in food pickup and increasing the chances of spoilage. Since food is perishable, timely communication is critical.

Another issue is the absence of efficient location-based matching. Donors and receivers are often not matched based on proximity, resulting in longer travel times and inefficient logistics, which discourages participation from volunteers and NGOs.

Additionally, most existing solutions have limited scalability. They are often designed for small communities or specific regions and cannot handle large-scale operations involving multiple users, locations, and high volumes of data. This restricts their usability in real-world scenarios where demand and supply are dynamic.

Many systems also lack user-friendly interfaces and accessibility, making them difficult to use for individuals who are not technically skilled. This reduces user engagement and limits the platform's reach.

Another important limitation is the lack of food safety and quality tracking mechanisms. Existing platforms rarely include features to monitor food freshness, expiry time, or handling conditions, which can pose health risks to recipients.

Furthermore, most platforms do not utilize advanced technologies such as AI for prediction or optimization, which could significantly improve efficiency in food distribution and demand forecasting.

3.1 Research Gap

From the analysis of existing systems, the following research gaps have been identified:

- Lack of real-time communication and instant notification systems
- Inefficient or missing location-based matching algorithms
- Absence of integrated food quality and expiry tracking
- Limited use of automation and intelligent decision-making (AI/ML)
- Poor scalability and performance for large user bases
- Lack of a centralized and user-friendly platform combining all features

Table 2: Limitations of Existing Systems and Research Gaps

Sr. No.	Limitation of Existing System	Research Gap Identified
1	Lack of real-time coordination between donors and receivers	Need for instant communication and notification systems
2	No efficient location-based matching	Requirement of proximity-based matching algorithms
3	Limited scalability for large-scale operations	Need for scalable and high-performance system architecture
4	Poor user interface and accessibility	Need for user-friendly and accessible platform design
5	No food safety or quality tracking	Integration of food freshness and expiry monitoring
6	Absence of advanced technologies (AI/ML)	Use of AI/ML for prediction, optimization, and automation
7	Fragmented or non-centralized platforms	Need for a centralized system combining all features

Chapter 4

Problem Statement and Objectives

4. Problem Statement and Objectives

4.1 Problem Statement

Food waste represents a critical global challenge where approximately one-third of all food produced is discarded while millions suffer from hunger. The existing food donation systems are inefficient, relying on manual coordination through phone calls and lacking real-time information sharing. This creates a significant gap between potential food donors such as restaurants, events, and households, and NGOs that work to feed the underprivileged. The absence of a centralized, accessible platform results in missed opportunities for food redistribution, leading to unnecessary waste and unfulfilled charitable potential. There is an urgent need for a technology-driven solution that can bridge this gap by providing real-time visibility of surplus food, enabling quick coordination between donors and NGOs, while ensuring trust and accountability through proper verification mechanisms.

4.2 Objectives

The primary objectives of the ResQFood project are as follows:

- To develop a web-based platform that connects food donors with NGOs in real-time, facilitating efficient food redistribution and reducing food waste.
- To implement a multi-role authentication system that verifies users based on organizational credentials such as Aadhar for individuals and registration IDs for NGOs, ensuring platform security and building trust.
- To create an intuitive user interface that allows donors to easily upload food details including quantity, type, preparation time, and pickup location with minimal effort.
- To provide NGOs with a searchable database of available food donations with filtering capabilities based on location, quantity, and food type for efficient matching.
- To enable real-time claiming functionality where NGOs can reserve food donations and coordinate pickup times directly through the platform.
- To design a responsive web application that works seamlessly across different devices including desktops, tablets, and smartphones for maximum accessibility.
- To implement data validation and quality control mechanisms that prevent spam listings and ensure the reliability of information on the platform.

  To contribute to environmental sustainability by reducing the amount of food that ends up in landfills and consequently decreasing greenhouse gas emissions from food waste

Chapter 5

Proposed System

5. Proposed System

The ResQFood system is designed as a comprehensive web-based solution that addresses the limitations of existing food donation platforms. The proposed system architecture follows a client-server model with a clear separation of concerns, utilizing modern web technologies to deliver a robust, scalable, and user-friendly platform.

5.1 System Architecture

The system is built on a three-tier architecture consisting of the presentation layer, application layer, and data layer. The presentation layer comprises HTML, CSS, and JavaScript files that provide the user interface and handle client-side interactions. The application layer, built using Django framework, manages business logic, user authentication, and API endpoints. The data layer utilizes a relational database to store user information, food listings, and transaction records.

The architecture is organized into distinct modules for better maintainability and scalability. The core module contains project configuration and main routing logic. The apps directory houses business logic divided into two main components: the listings module that handles food surplus and claiming functionality, and the users module that manages multi-role authentication and user profiles. The static directory contains frontend assets including CSS for styling with a modern theme and JavaScript for UI transitions and API calls. The templates directory holds HTML files that serve as the main dashboard and authentication modals.

5.2 Custom User Model and Authentication

The system implements a custom user model that extends Django's built-in authentication to support multiple user types. Each user is assigned a role that determines their capabilities within the system. Donors, including restaurants, event organizers, and households, can create and manage food listings. NGOs can search for and claim available food donations. The authentication process verifies users based on their organizational credentials, with Aadhar verification for individual donors and registration ID verification for NGOs.

5.3 Food Listing Management

The food listing module provides comprehensive functionality for managing food donations throughout their lifecycle. Donors can create new listings by providing essential information such as food type, quantity, preparation time, expiry date, and pickup location. The system validates this information to ensure data quality and completeness. Each listing is associated with the donor's user account through a foreign key relationship, enabling tracking and accountability.

The listing interface presents information in a clear, organized manner with status indicators showing whether food is available, claimed, or collected. Donors can update listing details or mark items as no longer available if circumstances change. The system automatically manages listing visibility based on status, ensuring that NGOs only see available items when searching.

5.4 Search and Claiming Functionality

NGOs access a dedicated search interface that displays all available food donations in an organized format. The search functionality includes multiple filters allowing NGOs to narrow down results based on their specific needs and capacity. Location-based filtering helps NGOs identify donations in their operational area, while quantity filters ensure they can handle

the volume of food being offered. Food type filters enable organizations to match donations with the dietary requirements or preferences of the communities they serve.

The claiming process is designed to be quick and straightforward. When an NGO identifies a suitable donation, they can claim it with a single action. The system immediately updates the listing status and notifies the donor of the claim. This real-time coordination minimizes the time between food availability and collection, which is critical for perishable items. The system also prevents double-claiming by locking listings once they are claimed by an NGO.

5.5 User Interface Design

The user interface follows modern design principles with a focus on clarity, simplicity, and ease of use. The design incorporates a contemporary theme with smooth animations and transitions that provide visual feedback for user actions. The color scheme is carefully chosen to be appealing while maintaining good contrast for readability. The layout is responsive, automatically adjusting to different screen sizes to ensure a consistent experience whether accessed from a desktop computer, tablet, or smartphone.

Navigation is intuitive with clear labels and logical organization of features. The main dashboard provides quick access to all key functions based on user role. Donors see options to create new listings and view their donation history, while NGOs see the search interface and their claimed items. Authentication is handled through modal dialogs that overlay the main interface, providing a seamless login and signup experience without navigating away from the current page.

5.6 Data Model and Relationships

The database design employs a relational model with well-defined relationships between entities. The User table stores authentication credentials and profile information for both donors and NGOs, with role designation determining access permissions. The FoodItem table contains details about food listings and maintains a foreign key relationship with the User table to link each listing to its donor. This relationship ensures that donors can only modify or delete their own listings while enabling the system to track donation patterns and generate statistics.

Additional fields in the FoodItem table track the claiming NGO and timestamps for listing creation, claiming, and collection. This temporal data is valuable for analytics and helps both donors and NGOs understand patterns in food availability and collection efficiency. The

database design also supports future enhancements such as rating systems or detailed reporting features without requiring major structural changes.

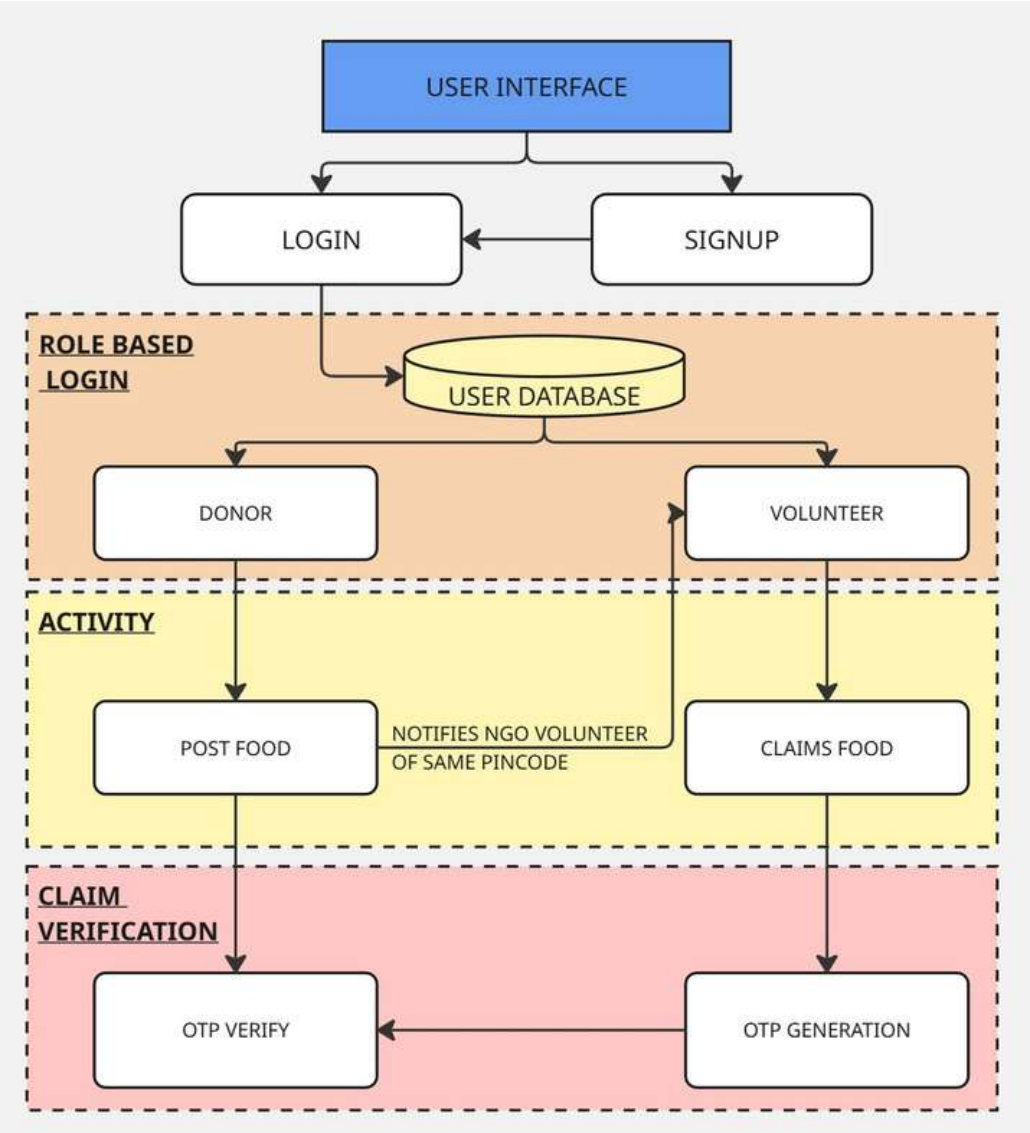

Figure 1 : Workflow

Chapter 6

Experimental Setup

6. Experimental Setup

6.1 Development Environment

The development of ResQFood was carried out using industry-standard tools and technologies. It was primarily developed on Windows systems, though the chosen stack ensures cross-platform compatibility. Visual Studio Code was used as the main IDE, offering features like syntax highlighting, debugging, and version control integration to enhance productivity.

6.2 Software Requirements

The software stack for ResQFood consists of carefully selected technologies that deliver a robust and scalable solution. Python 3.x serves as the primary programming language, chosen for its simplicity and extensive library ecosystem. Django framework version 4.x was selected

as the backend due to its batteries-included philosophy, built-in authentication system, and strong database support.

For the database layer, SQLite was used during development for its zero-configuration setup and portability, with the system designed to easily migrate to PostgreSQL or MySQL for production deployment. The frontend utilizes HTML5 for structure, CSS3 for styling with custom properties for theming, and vanilla JavaScript for dynamic interactions and API calls. This choice of vanilla JavaScript over heavy frameworks keeps the application lightweight and ensures fast load times.

6.3 Hardware Requirements

The hardware requirements for developing and testing ResQFood are minimal, making the platform accessible for deployment in various environments. For development purposes, a standard laptop or desktop computer with at least 4GB RAM and a dual-core processor is sufficient. The server hosting the application requires similar modest specifications for small to medium-scale deployments, with the ability to scale up as user base grows.

Client-side hardware requirements are equally undemanding, with any device capable of running a modern web browser being able to access the platform. This includes desktop computers, laptops, tablets, and smartphones manufactured in the last five years. The responsive design ensures optimal display across different screen sizes and resolutions.

6.4 Database Configuration

The database setup involves creating a schema that efficiently stores and retrieves information about users, food listings, and transactions. During development, Django's migration system was used to manage database schema changes incrementally. Initial data seeding included sample donor and NGO accounts for testing purposes, along with various food listing examples that represent different scenarios such as restaurant surplus, event leftovers, and household donations.

6.5 Testing Environment

Testing was conducted across multiple browsers including Google Chrome, Mozilla Firefox, and Microsoft Edge to ensure cross-browser compatibility. Mobile responsiveness was verified using browser developer tools and actual testing on Android and iOS devices with different screen sizes. The application was tested under various network conditions including

high-speed broadband and slower mobile connections to ensure acceptable performance across different connectivity scenarios.

6.6 Performance Evaluation Parameters

The system's performance was evaluated based on several key metrics. Response time was measured for critical operations such as user authentication, food listing creation, and search queries. Page load times were monitored to ensure that the application remains responsive even with a growing number of listings. Database query performance was analyzed to identify and optimize any bottlenecks. User experience metrics included ease of completing common tasks and overall satisfaction with the interface design and functionality.

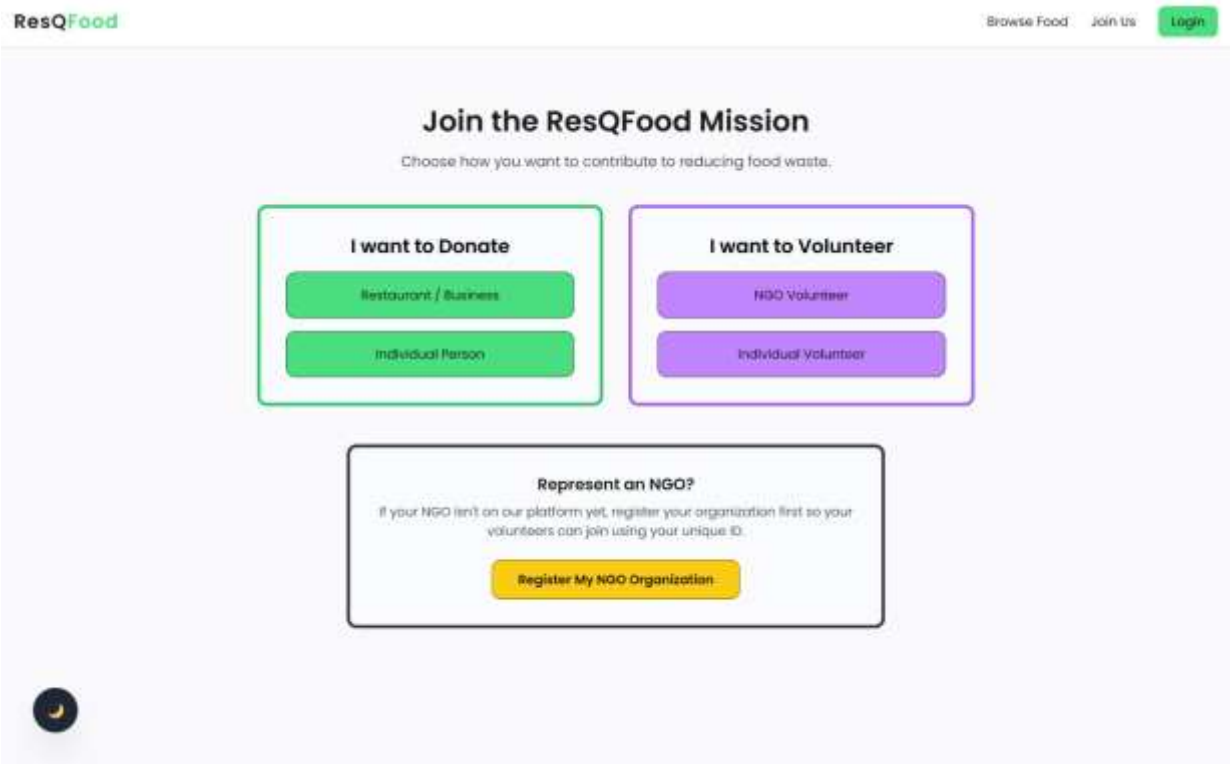

Figure 2 : Sign-Up

ResQFood sign-up options for different roles.

Figure 3 : Donor Sign-Up Form

Sign-Up Forms for different donors

Figure 4 : Volunteer Sign-Up Forms

Sign-Up Forms for different volunteers

Figure 5 : NGO Registration Form

Registration Form for NGOs

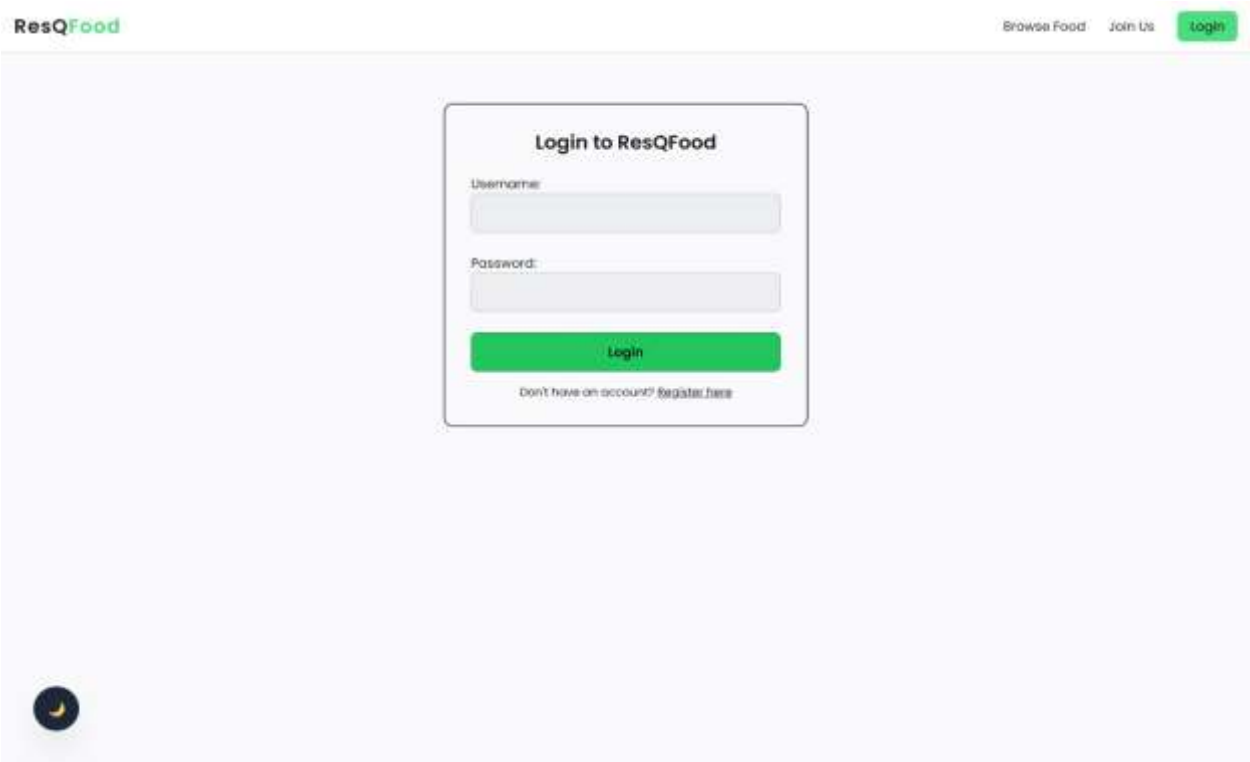

Figure 6 : Login

Login Page for Users

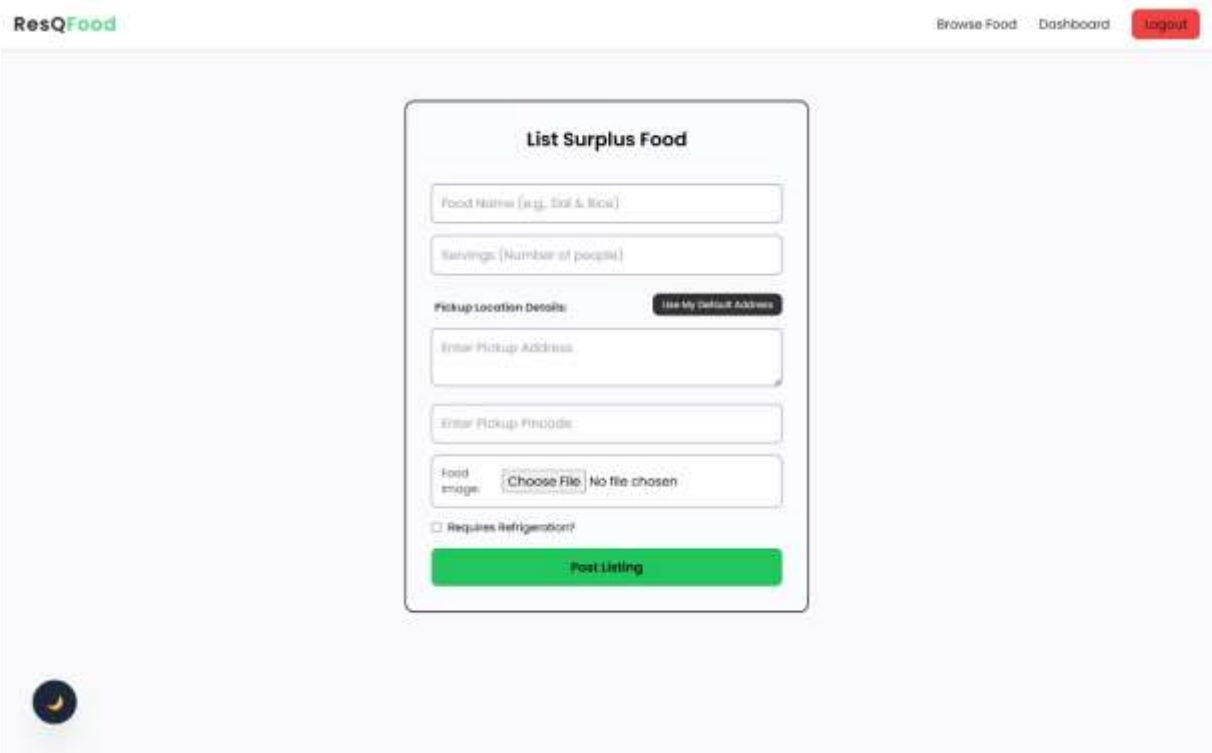

Figure 7 : Food Listing Form

Food Listing Page to list surplus food

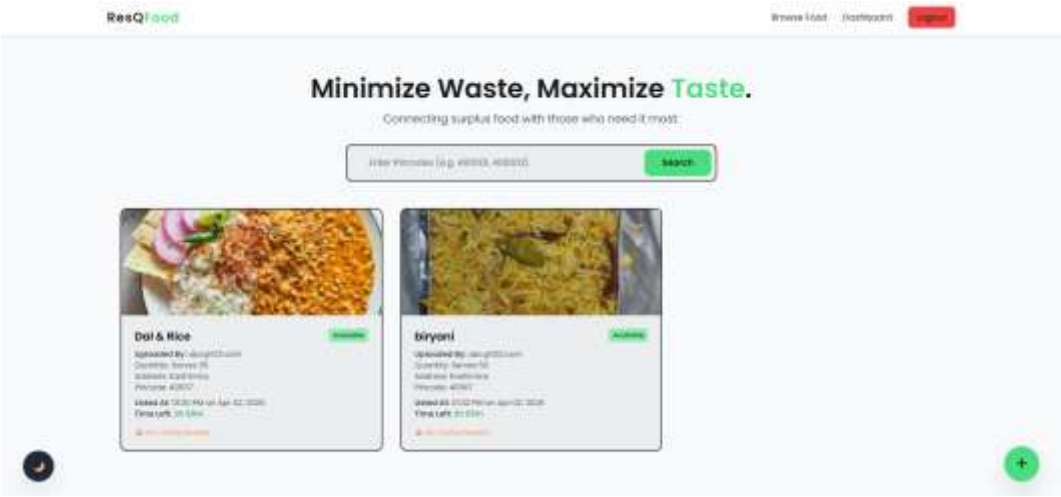

Figure 8 : Listed Food

Surplus Food Listing Page (Light Mode)

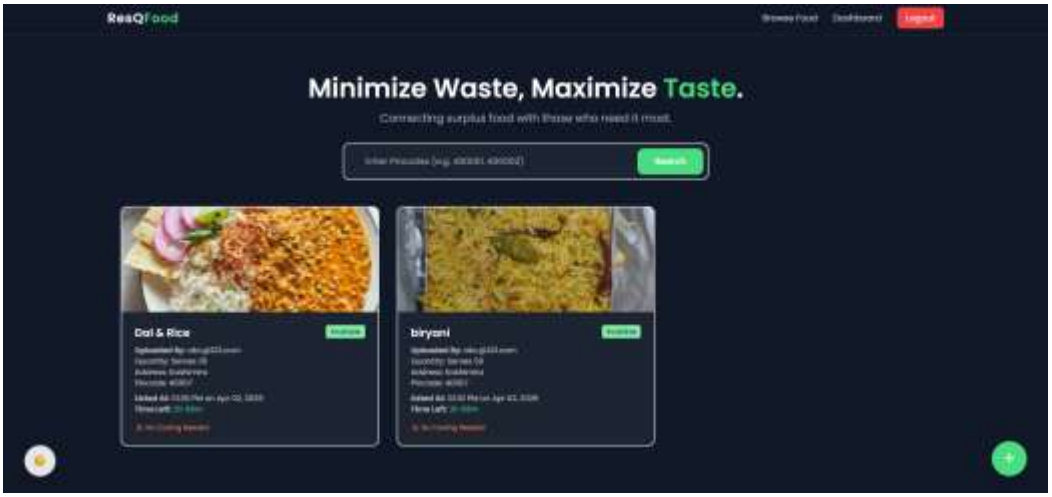

Figure 9 : Food Listings

Surplus Food Listing Page (Dark Mode)

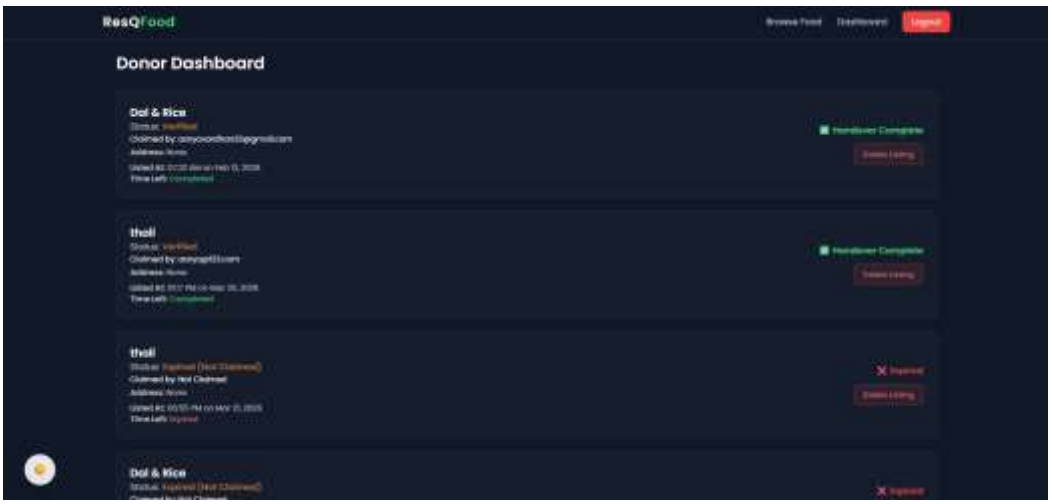

Figure 10 : Dashboard

Dashboard to view Past orders

Chapter 7

Results and Discussion

7. Results and Discussion:

7.1 Results:

- The ResqFood system was successfully designed and implemented as a web-based platform aimed at reducing food wastage and improving food distribution efficiency. The system connects hotels, NGOs, and volunteers through a role-based interface and ensures secure food pickup using a One-Time Password (OTP) verification mechanism.
- The system effectively enables hotels to list surplus food and allows NGOs and volunteers to search and claim food based on their pincode. Once the food is claimed, a 4-digit OTP is generated, which ensures secure and verified pickup at the time of collection.
- The application was tested under various scenarios, including user registration, login authentication, food listing, searching, claiming, and OTP-based pickup verification. All modules performed reliably, and the workflow remained consistent without errors.

- The system is developed using a backend framework integrated with a database, enabling real-time data storage, retrieval, and multi-user synchronization. The backend handles user authentication, role-based access control, and secure transaction processing, while the database ensures data integrity, consistency, and scalability.
- The proposed system simplifies the food redistribution process and reduces dependency on manual coordination. It enhances transparency, ensures accountability, and minimizes food wastage. The OTP-based verification mechanism further strengthens system security by ensuring that food is collected only by authorized users.
- Overall, the ResqFood system presents a practical, efficient, and scalable solution to the problem of food wastage and hunger. With future enhancements such as mobile application development, real-time notifications, and advanced tracking systems, the platform can be further improved to maximize its impact

7.2 Discussion:

- The ResqFood system was successfully developed as a web-based application aimed at minimizing food wastage by connecting food donors such as hotels and individuals with NGOs and volunteers. The system incorporates role-based access, location-based search using pincode, and a secure OTP verification mechanism to ensure proper and efficient food distribution.
- During implementation, the system was tested across multiple modules, including user registration, login authentication, role selection, food listing, food searching, and pickup verification. Each module was evaluated for functionality, usability, and accuracy, and the results indicate that the system performs efficiently and meets the intended objectives.
- In the hotel module, users were able to successfully list leftover food by providing g details such as food description, quantity, and pincode. The listed food was stored and displayed correctly in the system. Additionally, only verified users were allowed to perform critical actions such as listing food, thereby improving system reliability and trust.
- In the NGO and volunteer module, users were able to search for available food using pincode-based filtering. This feature enabled users to quickly locate nearby food resources. Once food was claimed, the system generated a unique 4-digit OTP, which served as a secure verification code during the pickup process.
- The OTP verification mechanism worked effectively in all test cases. When the correct OTP was entered by the hotel, the system marked the transaction as completed. Incorrect

OTP entries were rejected, preventing unauthorized pickups and enhancing system security and accountability.

- The system demonstrated efficient performance with fast response times and smooth navigation. The user interface was designed to be simple, interactive, and user-friendly, allowing users with minimal technical knowledge to operate it easily. The dashboards provided clear and structured information regarding food listings, pickups, and status updates.
- The results indicate that the proposed system significantly reduces the gap between food surplus and food demand. It eliminates the need for manual coordination and provides a structured, efficient, and transparent method for food redistribution. The role-based system further improves organization by ensuring users interact only with relevant functionalities.
- However, certain limitations were observed during testing. The system currently relies on a client-side database, which limits scalability and real-time synchronization across multiple users. Additionally, advanced features such as live tracking, real-time notifications, and dynamic updates are not included in the current prototype.
- Despite these limitations, the system provides a strong foundation for future development. By integrating a backend database such as Firebase or MySQL, the system can be scaled for real-time operations. Additional features like GPS-based tracking, push notifications, and mobile application integration can further enhance system performance and usability.

In conclusion, the ResqFood system is a practical, efficient, and secure solution for reducing food wastage and improving food redistribution. It demonstrates how technology can be effectively used to address real-world social challenges and has strong potential to evolve into a full-scale, impactful application.

Chapter 8

Conclusion and Future Work

8. Conclusion and Future Work

8.1 Conclusion

In today's world, food wastage is a major issue that impacts both the environment and society, especially in urban areas where large amounts of food are discarded while many people still lack access to proper meals. The ResQFood system is designed as a practical solution by providing a platform where donors like restaurants, event organizers, and households can share details of surplus food, and NGOs can quickly claim it. This real-time interaction helps reduce delays and improves efficiency.

The system is user-friendly and easy to operate, even for individuals with basic technical knowledge, and includes basic verification features to build trust among users. Overall, the project demonstrates how technology can effectively address real-world problems by reducing food wastage, supporting social welfare, and encouraging responsible food distribution.

Although currently developed on a small scale, it has strong potential for expansion and wider implementation in the future.

8.2 Future Work

While the current system provides a basic and functional solution, there are several areas for improvement to enhance its performance and usability. Developing a mobile application can make the platform more accessible and convenient, allowing users to receive instant notifications and updates. Adding GPS-based tracking can help NGOs easily locate donors and plan efficient collection routes, reducing coordination time.

The system can also be improved by implementing advanced verification methods, such as government ID or NGO registration validation, to increase security and trust among users. Additionally, integrating Artificial Intelligence (AI) can help analyze data, predict food demand, and suggest optimal matches between donors and NGOs. Integration with government or municipal systems, along with awareness campaigns and partnerships, can further expand the platform's reach. Overall, with these improvements, ResQFood has strong potential to become a powerful solution for reducing food wastage and supporting social welfare on a larger scale.

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